

S T U D E N T W O R K B O O K

GET 311

Engineering Mathematics III

Key ideas, application questions and space to work

8 topics · 8 engineering departments · 128 questions

THIS WORKBOOK BELONGS TO

Name

Matriculation No.

Department

Session

3 Units · Compulsory · 45 Lecture Hours

How to use this workbook

This workbook accompanies the GET 311 lectures. Each of the eight topics opens with a short summary of the key ideas, followed by sixteen application questions: two for each engineering department that takes the course. Every question has space for your working, and each topic ends with a short review.

1	Read	Go through the key ideas for a topic before the lecture on it. Underline anything you do not follow.
2	Attempt	Answer the two questions for your own department first, then try the others. Most use the same method in a different setting.
3	Show your working	Use the ruled space under each question. Write every step so a tutor can see where a method went wrong.
4	Review	Finish each topic by rating your confidence and noting questions for your lecturer or tutor.

LEARNING OUTCOMES

By the end of the course you should be able to:

1. Show a clear understanding of the course content, with broad knowledge of the areas covered.
2. Have a deep enough foundation to go on to advanced mathematical topics.
3. Develop simple algorithms and compute accurately.
4. Write simple proofs of theorems and apply them.
5. Communicate mathematical knowledge well in speech, in writing and in groups.

DEPARTMENT CODES

Each question is tagged with the department it is written for:

AGE	Agricultural Engineering	CHE	Chemical Engineering
CPE	Computer Engineering	CVE	Civil Engineering
EEE	Electrical and Electronics Engineering	FDE	Food Engineering
MEE	Mechanical Engineering	PTE	Petroleum Engineering

Course map and progress tracker

Tick off your progress as you work through the course. Aim to attempt at least your own department's questions in every topic.

	Topic	Questions done	Confidence (1–4)	Tutor initials
1	Linear Algebra	___ / 16	1 2 3 4	
2	Elements of Matrices, Determinants and Inverses of Matrices	___ / 16	1 2 3 4	
3	Theory of Linear Equations; Eigenvalues and Eigenvectors	___ / 16	1 2 3 4	
4	Analytical Geometry, Coordinate Transformation, Solid Geometry, and Polar, Cylindrical and Spherical Coordinates	___ / 16	1 2 3 4	
5	Elements of Functions of Several Variables; Surface Variables	___ / 16	1 2 3 4	
6	Ordinary Integrals; Double, Triple, Line and Surface Integrals; Vector Calculus	___ / 16	1 2 3 4	
7	Gauss's, Green's and Stokes' Theorems and Applications; Single-Valued and Multivalued Functions	___ / 16	1 2 3 4	
8	Analytic Functions, Cauchy–Riemann Equations, Singularities and Zeros, Contour Integration and Bilinear Transformation	___ / 16	1 2 3 4	

T O P I C 1

Linear Algebra

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

Linear algebra studies **vectors**, **vector spaces** and **linear transformations**. A vector space is a set of objects that can be added together and multiplied by scalars under a fixed set of rules (axioms). Examples are $\mathbb{R}^n$, the set of polynomials and the set of solutions of a linear differential equation. The main ideas are:

- ▶ **Linear combination and span:** the set of all vectors you can build from a given set by scaling and adding.
- ▶ **Linear independence:** no vector in the set can be written as a combination of the others.
- ▶ **Basis and dimension:** a basis is a minimal spanning set, and the number of vectors in it is the dimension of the space.
- ▶ **Linear transformations:** maps T with $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$. Every such map between finite-dimensional spaces can be written as a matrix.
- ▶ **Inner product, norm and orthogonality:** tools for measuring length and angle, projecting and finding least-squares approximations.

Engineering uses linear algebra constantly. Structural analysis, circuit networks, signal processing, computer graphics, process mass balances and reservoir simulation all reduce to linear models, which makes this topic the base for the rest of the course.

MY NOTES

Application questions

1.01
AGE

A fertilizer blend is made from three stock fertilizers with N-P-K compositions $\mathbf{a} = (20, 10, 10)$, $\mathbf{b} = (0, 20, 20)$ and $\mathbf{c} = (10, 0, 30)$ (percent by mass). Decide whether these three vectors are linearly independent. Can any target composition in $\mathbb{R}^3$ be reached by blending them? Explain using the idea of span.

Working

1.02
AGE

Soil samples from a farm are described by the vector (moisture %, organic matter %, pH). Given $\mathbf{s}_1 = (22, 3.1, 6.2)$, $\mathbf{s}_2 = (18, 2.4, 5.8)$ and a reference sample $\mathbf{r} = (20, 2.8, 6.0)$, find the Euclidean distance of each sample from the reference and the angle between $\mathbf{s}_1$ and $\mathbf{s}_2$. Comment on how alike the samples are.

Working

1.03
CHE

The combustion of methane, $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$, can be balanced by writing the atom counts (C, H, O) of each species as vectors and finding a non-trivial vector in the null space of the resulting matrix. Set up the vector equation and find the smallest integer stoichiometric coefficients.

Working

1.04
CHE

In a three-component mixture, the composition vectors of streams A, B and C are $\mathbf{x}_A = (0.5, 0.3, 0.2)$, $\mathbf{x}_B = (0.2, 0.5, 0.3)$ and $\mathbf{x}_C = (0.3, 0.2, 0.5)$. Show that the set $\{\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C\}$ is a basis of $\mathbb{R}^3$, and express a product stream $\mathbf{x}_P = (0.35, 0.35, 0.30)$ as a linear combination of them.

Working

1.05
C P E

A 2-D graphics engine rotates every point by $\theta = 30^\circ$ and then scales it by 2 in the x-direction. Write each step as a linear transformation matrix, find the combined matrix, and apply it to the vertices of the unit square.

Working

1.06
C P E

In a basic document-search system, each document is a term-frequency vector. With $\mathbf{d}_1 = (3, 0, 1, 2)$ and $\mathbf{d}_2 = (1, 1, 0, 2)$, compute the cosine similarity between them. Write a short algorithm (pseudocode) that returns the document most similar to a query vector $\mathbf{q}$.

Working

1.07
C V E

Three forces act at a truss joint: $\mathbf{F}_1 = (10, 0)$ kN, $\mathbf{F}_2 = (-5, 8)$ kN and $\mathbf{F}_3 = (a, b)$ kN. Use vector addition to find $\mathbf{F}_3$ so the joint is in equilibrium. Are $\mathbf{F}_1$ and $\mathbf{F}_2$ linearly independent? What does that mean for the joint?

Working

1.08
C V E

A surveyor records displacement vectors between control points: $\mathbf{AB} = (120, 45, 2)$ m and $\mathbf{BC} = (-30, 80, -1)$ m. Find $\mathbf{AC}$, its length, and the unit vector along $\mathbf{AC}$. Find the projection of $\mathbf{BC}$ on $\mathbf{AB}$.

Working

1.09

EEE

A three-phase balanced voltage set can be written as vectors in $\mathbb{R}^2$ (phasor components). Show that the phasors $V\angle 0^\circ$, $V\angle -120^\circ$ and $V\angle 120^\circ$ are linearly dependent, and that their sum is the zero vector.

Working

1.10

EEE

A discrete signal of 4 samples, $\mathbf{x} = (4, 2, 0, 2)$, is expanded in the orthogonal basis $\mathbf{h}_1 = (1, 1, 1, 1)$, $\mathbf{h}_2 = (1, 1, -1, -1)$, $\mathbf{h}_3 = (1, -1, 0, 0)$, $\mathbf{h}_4 = (0, 0, 1, -1)$ (a Haar-type basis). Check that the basis is orthogonal, then find the coefficients of $\mathbf{x}$ using inner products.

Working

1.11

FDE

A food product is formulated from three ingredients whose (protein, fat, carbohydrate) contents in g per 100 g are $(12, 2, 70)$, $(25, 15, 5)$ and $(1, 0, 95)$. Decide whether any nutritional profile can be made by mixing them (span). Find the mass of each ingredient needed to supply exactly $(10, 3, 75)$ g of protein, fat and carbohydrate.

Working

1.12

FDE

The colour of a juice is measured in (L, a, b^*) colour space. The target is $(45, 20, 30)$ and a batch measures $(42, 24, 27)$. Compute the colour difference ΔE as a Euclidean norm, and the unit vector giving the direction of the colour error.

Working

1.13
MEE

A force $\mathbf{F} = (3, -2, 5)$ N acts at the point $\mathbf{r} = (1, 2, 0)$ m. Compute the moment $\mathbf{M} = \mathbf{r} \times \mathbf{F}$ about the origin, and check that $\mathbf{M}$ is orthogonal to both $\mathbf{r}$ and $\mathbf{F}$ using the dot product.

Working

1.14
MEE

The displacement field of a deformed plate is modelled by a linear map $T(x, y) = (x + 0.02y, 0.01x + y)$. Write the matrix of T , find the image of the unit vectors, and decide whether T is invertible.

Working

1.15
PTE

The composition of a reservoir fluid is recorded as mole fractions of (C_1, C_2, C_3, C_4). Samples from wells W_1 and W_2 are (0.70, 0.12, 0.08, 0.10) and (0.60, 0.15, 0.10, 0.15). Find the composition of a blend with 40 % of W_1 and 60 % of W_2 , and the Euclidean distance between the two well compositions.

Working

1.16
PTE

A deviated well path is given by survey displacement vectors $\mathbf{d}_1 = (0, 0, -500)$ m, $\mathbf{d}_2 = (150, 100, -400)$ m and $\mathbf{d}_3 = (300, 200, -200)$ m. Find the total displacement vector to the target and the horizontal departure. Decide whether the three vectors are linearly independent.

Working

TOPIC 1 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

T O P I C 2

Elements of Matrices, Determinants and Inverses of Matrices

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

A **matrix** is a rectangular array of numbers set out in rows and columns. It is a compact way to store data and to represent linear transformations. The key ideas are:

- ▶ **Types of matrices:** square, diagonal, identity, triangular, symmetric, skew-symmetric, orthogonal and singular.
- ▶ **Matrix algebra:** addition, scalar multiplication, matrix multiplication (associative but *not* commutative) and the transpose.
- ▶ **Determinant:** a scalar attached to a square matrix, found by cofactor expansion or row reduction. It measures how much the transformation scales volume. A matrix is invertible exactly when $\det(A) \neq 0$.
- ▶ **Properties of determinants:** the effect of row operations, $\det(AB) = \det(A)\det(B)$, and $\det(A^T) = \det(A)$.
- ▶ **Inverse of a matrix:** A^{-1} satisfies $AA^{-1} = I$. It can be found with the adjoint formula $A^{-1} = \text{adj}(A)/\det(A)$ or with Gauss–Jordan elimination.
- ▶ **Rank:** the number of linearly independent rows or columns.

Engineers use matrices to store stiffness, conductance, transition and input–output data, and use inverses and determinants to solve for unknowns, test stability and change coordinates.

MY NOTES

Application questions

2.01
AGE

A farm makes three crops (maize, cassava, yam) over two seasons. The yield matrix Y (tonnes/ha) is 2×3 and the price vector p (₦/tonne) is 3×1 . With $Y = \begin{bmatrix} 2.5, 12, 8 \\ 3.0, 10, 9 \end{bmatrix}$ and $p = (150\,000, 40\,000, 60\,000)^T$, compute the revenue per hectare for each season using matrix multiplication.

Working

2.02
AGE

A tractor's hitch linkage gives the transformation matrix $A = \begin{bmatrix} 2, 1 \\ 1, 3 \end{bmatrix}$ between input and output displacements. Find $\det(A)$ and A^{-1} , and explain what $\det(A)$ says about how the linkage changes the area of the motion region.

Working

2.03
CHE

A mixer takes three feed streams, and the composition matrix C (rows = components, columns = streams) is $\begin{bmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.3 & 0.7 \end{bmatrix}$. Find C^{-1} and use it to get the feed flow rates that give component product flows (50, 40, 60) kg/h.

Working

2.04
CHE

The rate constants of a reversible reaction network are placed in a 3×3 matrix K . Show that if the rows of K add to zero (mass conservation), then $\det(K) = 0$. Explain why this makes K singular in physical terms.

Working

2.05
C P E

In a Hill cipher, a message is encrypted with the key matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \pmod{26}$. Find $\det(K) \pmod{26}$, check that K is invertible mod 26, and find the decryption matrix $K^{-1} \pmod{26}$.

Working

2.06
C P E

Homogeneous coordinates use 3×3 matrices for 2-D translation, rotation and scaling. Write the matrix that translates by $(4, -2)$ and then rotates by 90° . Find its determinant and inverse, and explain why the determinant is 1.

Working

2.07
C V E

A two-member spring model of a structure has stiffness matrix $K = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$ with $k_1 = 200 \text{ kN/m}$ and $k_2 = 150 \text{ kN/m}$. Find K^{-1} (the flexibility matrix) and the displacements under loads $F = (10, 5)^T \text{ kN}$.

Working

2.08
C V E

Show that the stiffness matrix of an unsupported beam element has determinant zero. Explain, using rank, why supports (boundary conditions) must be applied before the system can be solved.

Working

2.09
EEE

A two-port network has the impedance matrix $Z = \begin{bmatrix} 10 & 4 \\ 4 & 6 \end{bmatrix} \Omega$. Find the admittance matrix $Y = Z^{-1}$ and the port currents when $V = (20, 12)^T V$.

Working

2.10
EEE

Nodal analysis of a three-node resistive circuit gives the conductance matrix $G = \begin{bmatrix} 0.5 & -0.2 & 0 \\ -0.2 & 0.7 & -0.3 \\ 0 & -0.3 & 0.4 \end{bmatrix} S$. Compute $\det(G)$ by cofactor expansion and state whether a unique set of node voltages exists.

Working

2.11

F D E

A bakery makes bread, cake and biscuits. The ingredient-use matrix (kg of flour, sugar and fat per batch) is $A = \begin{bmatrix} 5 & 3 & 2 \\ 0.5 & 2 & 1 \\ 0.3 & 1.5 & 1 \end{bmatrix}$. With daily stock (80, 25, 18) kg, find A^{-1} and the number of batches that uses all the stock exactly.

Working

2.12

F D E

The heat-transfer coefficients between three zones of a tunnel dryer form a symmetric matrix H . Given $H = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix}$, show that H is symmetric, compute $\det(H)$, and find H^{-1} .

Working

2.13
MEE

The rotation matrix about the z-axis is $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Prove that R is orthogonal ($R^T R = I$), that $\det R = 1$, and that $R^{-1} = R(-\theta)$.

Working

2.14
MEE

A two-degree-of-freedom spring-mass system has mass matrix $M = \text{diag}(2, 1)$ kg and stiffness matrix $K = \begin{bmatrix} 300 & -100 \\ -100 & 100 \end{bmatrix}$ N/m. Compute $M^{-1}K$, which is needed for the vibration analysis in Topic 3.

Working

2.15

PTE

A finite-difference model of pressure in a 1-D reservoir with three blocks gives the transmissibility matrix $T = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. Find T^{-1} and the block pressures when $T\mathbf{p} = (1500, 0, 1200)^T$ (suitably scaled).

Working

2.16

PTE

Three wells produce oil, gas and water. The production-rate matrix is $P = \begin{bmatrix} 500 & 200 & 50 \\ 300 & 400 & 100 \\ 200 & 100 & 300 \end{bmatrix}$ (units/day). Find $\det(P)$ and explain what a zero determinant would mean for back-calculating each well's contribution from total separator readings.

Working

TOPIC 2 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

TOPIC 3

Theory of Linear Equations; Eigenvalues and Eigenvectors

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

Systems of linear equations $Ax = b$ are the most common computational task in engineering. The theory answers three questions:

- ▶ **Existence and uniqueness (consistency):** by the Rouché–Capelli theorem, a system has a solution if and only if $\text{rank}(A) = \text{rank}([A \mid b])$. The solution is unique when that rank equals the number of unknowns. Otherwise there are infinitely many solutions.
- ▶ **Homogeneous systems:** $Ax = 0$ always has the trivial solution, and has non-trivial solutions exactly when $\det(A) = 0$.
- ▶ **Solution methods:** Cramer's rule, Gaussian and Gauss–Jordan elimination, LU decomposition, and iterative methods (Jacobi, Gauss–Seidel) for large sparse systems.

Eigenvalues and eigenvectors describe directions that a matrix only stretches: $Av = \lambda v$. The eigenvalues are the roots of the **characteristic equation** $\det(A - \lambda I) = 0$. Important results include:

- ▶ the sum of the eigenvalues is $\text{trace}(A)$ and their product is $\det(A)$;
- ▶ a symmetric matrix has real eigenvalues and orthogonal eigenvectors;
- ▶ **diagonalisation** $A = PDP^{-1}$, which makes powers and exponentials of A easy to compute;
- ▶ the **Cayley–Hamilton theorem**: every square matrix satisfies its own characteristic equation.

Eigen-analysis gives natural frequencies and vibration modes, the stability of dynamic systems, principal stresses, the steady states of Markov processes and the principal components of data.

MY NOTES

Application questions

3.01
AGE

An irrigation network has three canals with flows q_1, q_2, q_3 (m^3/s) that satisfy $q_1 + q_2 + q_3 = 12$, $2q_1 - q_2 = 3$ and $q_2 - q_3 = 1$. Use Gaussian elimination to find the flows, and check the result with Cramer's rule.

Working

3.02
AGE

A livestock population has calves, yearlings and adults. It is modelled by the Leslie matrix $L = \begin{bmatrix} 0 & 0 & 0.8 \\ 0.6 & 0 & 0 \\ 0 & 0.9 & 0.9 \end{bmatrix}$. Find the dominant eigenvalue and its eigenvector. What do they say about long-term herd growth and the herd's age structure?

Working

3.03
CHE

A system of three CSTRs in series with recycle gives the linear equations $5C_1 - C_3 = 20$, $-4C_1 + 5C_2 = 0$ and $-4C_2 + 5C_3 = 0$ (concentrations in mol/L). Decide whether the system is consistent, then solve it using LU decomposition.

Working

3.04
CHE

Two first-order reactions $A \rightleftharpoons B$ are modelled by $dx/dt = Kx$ with $K = \begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix}$. Find the eigenvalues and eigenvectors of K , and use them to write the general solution. Interpret the zero eigenvalue as the equilibrium composition.

Working

3.05

C P E

Write an algorithm (pseudocode or Python) for Gauss–Seidel iteration, and use it for 3 iterations on $10x - y = 9$, $-x + 10y - 2z = 7$, $-2y + 10z = 6$. State the convergence condition (diagonal dominance) and check it.

Working

3.06

C P E

A simplified PageRank for 3 web pages uses the column-stochastic link matrix $M = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}$. Show that $\lambda = 1$ is an eigenvalue, and find the normalised eigenvector that gives the page ranks.

Working

3.07
C V E

The plane stress at a point is given by the tensor $\sigma = \begin{bmatrix} 80 & 30 \\ 30 & 20 \end{bmatrix}$ MPa. Find the principal stresses (eigenvalues) and principal directions (eigenvectors), and give the angle of the principal plane.

Working

3.08
C V E

A three-storey shear building has mass matrix $M = mI$ and stiffness matrix $K = k \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$. Set up the eigenvalue problem $(K - \omega^2 M)\varphi = 0$, and find the natural frequencies in terms of k/m .

Working

3.09
EEE

Mesh analysis of a circuit gives $12I_1 - 4I_2 = 24$, $-4I_1 + 10I_2 - 2I_3 = 0$ and $-2I_2 + 8I_3 = -6$. Solve for the mesh currents with Cramer's rule.

Working

3.10
EEE

The state-space model of an RLC circuit is $\dot{x} = Ax$ with $A = \begin{bmatrix} 0 & 1 \\ -1/(LC) & -R/L \end{bmatrix}$. For $R = 4 \Omega$, $L = 1 \text{ H}$ and $C = 0.25 \text{ F}$, find the eigenvalues of A . Classify the circuit as over-, critically or under-damped, and say whether it is stable.

Working

3.11

FDE

A juice blender mixes three concentrates to meet targets for sugar, acidity and vitamin C. This gives $0.1x + 0.2y + 0.15z = 15$, $0.02x + 0.01y + 0.03z = 2$ and $0.5x + 0.2y + 0.4z = 40$. Decide using ranks whether the targets can be met exactly, and solve the system.

Working

3.12

FDE

The moisture transfer between the surface, middle and core layers of a drying grain kernel is modelled by $dM/dt = AM$ with $A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$. Find the eigenvalues of A and explain how the smallest eigenvalue in magnitude controls the drying time.

Working

3.13
MEE

For the spring-mass system of Topic 2 ($M = \text{diag}(2, 1)$, $K = \begin{bmatrix} 300 & -100 \\ -100 & 100 \end{bmatrix}$), solve $\det(K - \omega^2 M) = 0$ to get the two natural frequencies, and find and sketch the corresponding mode shapes.

Working

3.14
MEE

Use the Cayley-Hamilton theorem to find A^{-1} and A^3 for the inertia-like matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Check A^{-1} by direct computation.

Working

3.15
PTE

The material balance for three reservoir compartments linked by transmissibility gives $3p_1 - p_2 = 4000$, $-p_1 + 3p_2 - p_3 = 1000$ and $-p_2 + 2p_3 = 2500$ (psi). Solve for the compartment pressures with Gauss–Jordan elimination.

Working

3.16
PTE

The permeability tensor of an anisotropic formation is $k = [[200, 50], [50, 100]]$ mD. Find the principal permeabilities and the directions of maximum and minimum permeability, and explain how these directions affect horizontal well placement.

Working

TOPIC 3 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

T O P I C 4

Analytical Geometry, Coordinate Transformation, Solid Geometry, and Polar, Cylindrical and Spherical Coordinates

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

Analytical geometry describes geometric objects with algebraic equations. In the plane this means straight lines, circles and the conic sections (parabola, ellipse, hyperbola), which all come from the general second-degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.

Coordinate transformation changes the reference frame:

- ▶ *translation* moves the origin: $x = X + h, y = Y + k$;
- ▶ *rotation* turns the axes through θ : $x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta$. Choosing $\tan 2\theta = 2h/(a - b)$ removes the xy term, which reduces a conic to standard form.

Solid (3-D) geometry covers direction cosines and ratios, the equations of lines and planes, the angle between planes, the distance from a point to a plane, and quadric surfaces (sphere, cylinder, cone, ellipsoid, paraboloid, hyperboloid).

Curvilinear coordinate systems make problems with symmetry much simpler:

System	Coordinates	Relation to Cartesian	Volume element
Polar (2-D)	(r, θ)	$x = r \cos \theta, y = r \sin \theta$	$dA = r \, dr \, d\theta$
Cylindrical	(r, θ, z)	$x = r \cos \theta, y = r \sin \theta, z = z$	$dV = r \, dr \, d\theta \, dz$
Spherical	(ρ, θ, φ)	$x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$	$dV = \rho^2 \sin \varphi \, d\rho \, d\theta \, d\varphi$

Cylindrical coordinates suit pipes, shafts and wells. Spherical coordinates suit tanks, antennas and radial flow.

MY NOTES

Application questions

4.01
AGE

A centre-pivot irrigation system covers a circle of radius 400 m centred at (500, 300) m in the farm's coordinate grid. Write the equation of the irrigated boundary, convert it to polar coordinates centred at the pivot, and find the irrigated area.

Working

4.02
AGE

A grain silo is a cylinder of radius 3 m and height 10 m with a conical roof of height 2 m. Write the equations of both surfaces in cylindrical coordinates, and find the total storage volume.

Working

4.03

CHE

A spherical catalyst pellet of radius $R = 5$ mm has a reactant concentration $C(\rho)$ that depends only on the radial distance. Write the position of a point on the pellet surface in spherical coordinates, and explain why spherical coordinates reduce the 3-D diffusion problem to one variable.

Working

4.04

CHE

The cross-section of an elliptical storage tank is $9x^2 + 4y^2 - 36x + 24y + 36 = 0$. Use a translation of axes to reduce it to standard form, and find the centre, semi-axes and cross-sectional area.

Working

4.05

C P E

A robot arm's end-effector is located by $(r, \theta, z) = (0.8 \text{ m}, 60^\circ, 0.5 \text{ m})$ in cylindrical coordinates. Convert this to Cartesian coordinates, and write a function (pseudocode) that converts between Cartesian, cylindrical and spherical coordinates in both directions.

Working

4.06

C P E

A lidar sensor reports a point at range $\rho = 20 \text{ m}$, azimuth $\theta = 45^\circ$ and elevation angle 30° (so the polar angle is $\varphi = 60^\circ$). Convert it to Cartesian coordinates, then find the distance from the point to the ground plane $z = 0$ and to the plane $x + y + z = 10$.

Working

4.07

C V E

A road designer needs the equation of the plane that contains the points $A(0, 0, 100)$, $B(200, 0, 104)$ and $C(0, 150, 97)$ (metres) on a graded embankment. Find the plane equation, the direction cosines of its normal, and the angle it makes with the horizontal.

Working

4.08

C V E

The profile of a parabolic arch bridge passes through $(-30, 0)$, $(30, 0)$ and $(0, 15)$ m. Find its equation. Rotate the axes by 10° to model an inclined abutment, and write the transformed equation.

Working

4.09

EEE

The radiation pattern of a dipole antenna is proportional to $\sin \varphi$ in spherical coordinates. Convert the pattern surface $\rho = \sin \varphi$ to Cartesian form, identify it, and sketch it.

Working

4.10

EEE

A coaxial cable has an inner conductor of radius $a = 1$ mm and an outer conductor of radius $b = 4$ mm. Describe the dielectric region with inequalities in cylindrical coordinates, and find the cross-sectional area of the dielectric using polar coordinates.

Working

4.11

FDE

A cylindrical can with radius 4 cm and height 11 cm is heated during sterilisation. The slowest-heating point is on the axis at mid-height. Give its cylindrical coordinates relative to the centre of the base, and find its distance from the rim edge.

Working

4.12

FDE

An egg is approximated by the ellipsoid $x^2/4 + y^2/4 + z^2/9 = 1$ (cm). Find its volume, and use a coordinate transformation (stretching) to map it onto a unit sphere. Explain why this simplifies heat-transfer analysis.

Working

4.13

MEE

A cam profile is given in polar form by $r = 40 + 10 \cos \theta$ (mm). Convert it to Cartesian form, find the maximum and minimum radius (the follower lift), and find the area enclosed by the profile.

Working

4.14

MEE

Two machine members lie along the lines $(x - 1)/2 = (y + 1)/3 = z/6$ and $(x - 2)/3 = (y - 1)/(-2) = (z + 1)/6$. Find the angle between them and the shortest distance between the lines.

Working

4.15

PTE

Radial flow into a vertical well is analysed in cylindrical coordinates. The drainage area is a circle of radius $r_e = 300$ m around a well of radius $r_w = 0.1$ m. Write the flow domain in cylindrical coordinates, and find the volume of a reservoir 20 m thick with 25 % porosity.

Working

4.16

PTE

A directional well leaves the surface at $(0, 0, 0)$ and reaches a target at $(600, 800, -2500)$ m. Find the equation of the straight well path, its inclination from the vertical, and its azimuth. Express the target in spherical coordinates.

Working

TOPIC 4 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

T O P I C 5

Elements of Functions of Several Variables; Surface Variables

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

Most engineering quantities depend on more than one variable. Temperature changes with position and time, and pressure depends on volume and temperature. A **function of several variables** $f(x, y, \dots)$ assigns one value to each point of its domain. The main tools are:

- ▶ **Level curves and surfaces:** $f(x, y) = c$ (contours) and $f(x, y, z) = c$ (level surfaces, such as isotherms).
- ▶ **Limits and continuity** in more than one dimension.
- ▶ **Partial derivatives:** $\partial f / \partial x$, $\partial f / \partial y$ and higher-order derivatives. Clairaut's theorem says $f_{xy} = f_{yx}$.
- ▶ **Total differential:** $df = f_x dx + f_y dy$, used for error and sensitivity analysis.
- ▶ **Chain rule and implicit differentiation** for composite and implicitly defined functions.
- ▶ **Directional derivative and gradient:** $D_u f = \nabla f \cdot u$. The gradient points in the direction of steepest increase.
- ▶ **Taylor expansion** in two variables, used for linearisation.
- ▶ **Extrema:** stationary points classified by the second-derivative test ($D = f_{xx} f_{yy} - f_{xy}^2$), and constrained optimisation with **Lagrange multipliers**.

Surfaces and surface variables: a surface can be written explicitly as $z = f(x, y)$, implicitly as $F(x, y, z) = 0$, or parametrically as $\mathbf{r}(u, v)$. The parameters u and v are the *surface variables*. The tangent plane and normal come from ∇F or from $\mathbf{r}_u \times \mathbf{r}_v$, and the element of surface area is $dS = |\mathbf{r}_u \times \mathbf{r}_v| du dv$. These are needed for the surface integrals in Topic 6.

MY NOTES

Application questions

5.01
AGE

Crop yield (t/ha) is modelled by $Y(N, W) = 0.04N + 0.05W - 0.0001N^2 - 0.0002W^2 + 0.00005NW$, where N is nitrogen (kg/ha) and W is water (mm). Find the input levels that maximise yield, and confirm that the point is a maximum.

Working

5.02
AGE

The evapotranspiration rate $E = kT^{0.5} R^{0.8}$ depends on temperature T and radiation R . Use the total differential to estimate the percentage error in E when T is measured with 2 % error and R with 3 % error.

Working

5.03
CHE

For a van der Waals gas, $(P + a/V^2)(V - b) = RT$. Find $\partial P/\partial V$ at constant T and $\partial P/\partial T$ at constant V . Verify the cyclic relation $(\partial P/\partial V)_T(\partial V/\partial T)_P(\partial T/\partial P)_V = -1$.

Working

5.04
CHE

A cylindrical reactor of fixed volume 10 m^3 is to be built with the least surface area (and so the least material cost). Use Lagrange multipliers to find the best radius and height.

Working

5.05
C P E

Gradient descent minimises the loss $L(w_1, w_2) = (w_1 - 3)^2 + 2(w_2 + 1)^2 + w_1 w_2$. Compute ∇L , write the gradient-descent update rule, and carry out 3 iterations from $(0, 0)$ with learning rate 0.1.

Working

5.06
C P E

The brightness of a grayscale image is modelled as $I(x, y) = 100 e^{-(x^2 + y^2)/50}$. Compute the image gradient at $(3, 4)$, its magnitude (the edge strength) and its direction, and explain how edge-detection algorithms use this.

Working

5.07
C V E

The ground surface of a site is $z = 50 - 0.01x^2 - 0.02y^2$ (m). At the point (10, 20), find the slope in the direction of the unit vector $(3/5, 4/5)$, the direction of steepest descent (the drainage path), and the equation of the tangent plane.

Working

5.08
C V E

A rectangular beam cut from a circular log of diameter d is strongest in bending when bh^2 is largest. Use Lagrange multipliers to find b and h subject to $b^2 + h^2 = d^2$.

Working

5.09

EEE

The power dissipated in a resistor is $P = V^2/R$. Given $V = 220 \pm 2$ V and $R = 50 \pm 0.5$ Ω , use partial derivatives and the total differential to estimate the maximum absolute and relative error in P .

Working

5.10

EEE

The electric potential near a point charge is $V(x, y, z) = k/\sqrt{(x^2 + y^2 + z^2)}$. Show that V satisfies Laplace's equation $\nabla^2 V = 0$ away from the origin, and find the equation of the equipotential surface through $(1, 2, 2)$.

Working

5.11

FDE

During drying, the moisture content is $M(t, T) = M_0 e^{-k(T)t}$ with $k(T) = 0.01 e^{0.05(T - 60)}$. Find $\partial M / \partial t$ and $\partial M / \partial T$, and use the chain rule to find dM/dt when the dryer temperature rises as $T = 60 + 0.5t$.

Working

5.12

FDE

A rectangular box for packaging cereal must hold 3000 cm^3 . The top and bottom cost twice as much per cm^2 as the sides. Use Lagrange multipliers to find the dimensions that give the cheapest box.

Working

5.13

MEE

The temperature distribution in a plate is $T(x, y) = 100 - x^2 - 2y^2$ ($^{\circ}\text{C}$). A heat-seeking sensor at $(2, 1)$ moves in the direction of fastest heating. Find that direction and the rate of temperature change in it, and find the path of the sensor.

Working

5.14

MEE

The surface of a turbine blade is given parametrically by $\mathbf{r}(u, v) = (u \cos v, u \sin v, 0.5v)$ for $1 \leq u \leq 2$ and $0 \leq v \leq \pi/2$ (a helicoid). Find $\mathbf{r}_u \times \mathbf{r}_v$, the unit normal and the element of surface area dS .

Working

5.15

PTE

The pressure distribution around a producing well is $p(x, y) = p_e - (q\mu/(4\pi kh)) \ln[(x^2 + y^2)/r_e^2]$. Find ∇p , and show that the flow (along $-\nabla p$) is directed radially towards the well.

Working

5.16

PTE

The oil formation volume factor $B_o(p, T)$ is approximated by $B_o = 1.2 + 0.0001(T - 150) - 0.00002(p - 3000)$. Find the linear (Taylor) approximation of B_o about $(p, T) = (3000, 150)$, and estimate B_o at (2800 psi, 160 °F).

Working

TOPIC 5 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

TOPIC 6

Ordinary Integrals; Double, Triple, Line and Surface Integrals; Vector Calculus

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

This topic extends integration from a single variable to curves, regions, surfaces and volumes, and introduces the calculus of vector fields.

Integrals

- ▶ **Ordinary (single) integrals:** $\int f(x) dx$, reviewed along with improper integrals and differentiation under the integral sign.
- ▶ **Double integrals:** $\iint_R f(x, y) dA$, evaluated as iterated integrals, sometimes after changing the order of integration or converting to polar coordinates. They give area, mass, centroids and moments of inertia of plane laminae.
- ▶ **Triple integrals:** $\iiint_V f dV$, evaluated in Cartesian, cylindrical or spherical coordinates, using the Jacobian for a change of variables. They give volume, mass and centre of gravity of solids.
- ▶ **Line integrals:** $\int_C f ds$ and $\int_C \mathbf{F} \cdot d\mathbf{r}$ along a curve C . They give work done, circulation and path-dependence. A field is **conservative** (path-independent) when $\mathbf{F} = \nabla\phi$.
- ▶ **Surface integrals:** $\iint_S f dS$ and $\iint_S \mathbf{F} \cdot \mathbf{n} dS$. The second is the **flux** of $\mathbf{F}$ through S .

Vector calculus

- ▶ **Derivatives and integrals of vectors:** $d\mathbf{r}/dt$ gives velocity and acceleration along a path. Vector functions are integrated component by component.
- ▶ **Gradient $\nabla\phi$** of a scalar field: the direction and rate of greatest increase, normal to level surfaces.
- ▶ **Divergence $\nabla \cdot \mathbf{F}$:** the net outflow (source strength) per unit volume.
- ▶ **Curl $\nabla \times \mathbf{F}$:** the local rotation (circulation per unit area). A field is **irrotational** when $\nabla \times \mathbf{F} = 0$.
- ▶ **Flux of a vector field:** the rate at which a quantity (fluid, heat, charge) crosses a surface.

These are the building blocks of the governing equations of fluid mechanics, heat transfer, electromagnetism and elasticity.

Application questions

6.01
AGE

The rainfall intensity over a rectangular catchment $0 \leq x \leq 2$ km, $0 \leq y \leq 3$ km is $I(x, y) = 10 + 2x + y$ (mm/h). Use a double integral to find the total volumetric rainfall rate (m^3/h) over the catchment and the average intensity.

Working

6.02
AGE

Water flows through a soil profile with Darcy velocity field $\mathbf{q} = (-2x, -2y, 4z) \times 10^{-6}$ m/s. Compute $\nabla \cdot \mathbf{q}$ and $\nabla \times \mathbf{q}$. State whether the flow is incompressible and irrotational, and find a potential function if one exists.

Working

6.03
CHE

A cylindrical packed-bed reactor has radius 0.5 m and length 2 m. The catalyst density varies as $\rho(r) = 800(1 - 0.3r^2)$ kg/m³. Use a triple integral in cylindrical coordinates to find the total catalyst mass.

Working

6.04
CHE

The velocity in a pipe of radius R is $\mathbf{v} = v_{\max}(1 - r^2/R^2) \mathbf{k}$ (laminar Poiseuille flow). Compute the volumetric flow rate as the flux of $\mathbf{v}$ through the pipe cross-section, and show that the mean velocity is $v_{\max}/2$.

Working

6.05
CPE

A drone flies along $\mathbf{r}(t) = (10 \cos t, 10 \sin t, 2t)$ m for $0 \leq t \leq 2\pi$. Find its velocity, speed and acceleration, and the total path length. Write a short algorithm that approximates this arc length numerically with the trapezoidal rule, and compare the result with the exact value.

Working

6.06
CPE

A heat sink on a processor chip occupies the region $0 \leq x \leq 2$, $0 \leq y \leq 2$, $0 \leq z \leq 1$ (cm). The heat generation rate is $q(x, y, z) = 5xy + z$ (W/cm³). Use a triple integral to find the total heat generated, and write it as a nested-loop numerical approximation (a Riemann sum).

Working

6.07
C V E

A lamina shaped like a T-beam cross-section occupies the region between $y = 0$ and $y = 4$ for $-1 \leq x \leq 1$ (the web), plus the region $4 \leq y \leq 5$ for $-3 \leq x \leq 3$ (the flange). Use double integrals to find the centroid $\bar{y}$ and the second moment of area I_x about the centroidal axis.

Working

6.08
C V E

A crane lifts a load along a path in the force field $\mathbf{F} = (2xy, x^2 + z, y)$ kN. Compute the work done $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0, 0)$ to $(1, 2, 3)$ along two different paths. Decide whether $\mathbf{F}$ is conservative, and find its potential if it is.

Working

6.09

EEE

The electric field of an infinite line charge is $\mathbf{E} = (\lambda/(2\pi\epsilon_0 r)) \hat{\mathbf{r}}$. Compute the flux of $\mathbf{E}$ through a closed cylinder of radius a and length L coaxial with the line. Show that it equals $\lambda L/\epsilon_0$, as Gauss's law requires.

Working

6.10

EEE

The magnetic field around a long straight wire is $\mathbf{B} = (\mu_0 I/(2\pi r)) \hat{\boldsymbol{\theta}}$. Evaluate the line integral $\oint \mathbf{B} \cdot d\mathbf{r}$ around a circle of radius a centred on the wire, and compute $\nabla \times \mathbf{B}$ for $r \neq 0$. Comment on the results.

Working

6.11
FDE

A hemispherical dome-shaped cake of radius 10 cm has a temperature during baking of $T(\rho) = 180 - 60(1 - \rho/10)$ °C. Use a triple integral in spherical coordinates to find the volume-averaged temperature.

Working

6.12
FDE

Air flows through a rectangular dryer opening $0 \leq x \leq 1$ m, $0 \leq z \leq 0.5$ m with velocity field $\mathbf{v} = (0, 2 + x - z, 0)$ m/s. Compute the volumetric air flow rate (the flux through the opening). If the air carries 0.012 kg of water per m^3 , find the rate of moisture removal.

Working

6.13
MEE

Find the mass moment of inertia about the z -axis of a solid cone of height h , base radius a and uniform density ρ , using a triple integral in cylindrical coordinates.

Working

6.14
MEE

A fluid velocity field is $\mathbf{v} = (y, -x, 0)$. Compute $\nabla \cdot \mathbf{v}$ and $\nabla \times \mathbf{v}$, interpret the flow physically (rigid-body rotation), and evaluate the circulation around the unit circle in the xy -plane.

Working

6.15
 PTE

A reservoir occupies the region between the surfaces $z = -2000$ and $z = -2000 - 0.001(x^2 + y^2)$ for $x^2 + y^2 \leq 500^2$ (m). Use a triple integral in cylindrical coordinates to find the bulk rock volume. With 20 % porosity and 75 % oil saturation, estimate the oil in place.

Working

6.16
 PTE

The pressure field in a reservoir is $p(x, y) = 3000 - 0.5x^2 - 0.3y^2$ (psi), and the Darcy velocity is $\mathbf{u} = -(k/\mu)\nabla p$. Find $\mathbf{u}$, and compute the flux of $\mathbf{u}$ across the boundary of the square $-100 \leq x, y \leq 100$ m (per unit thickness) with $k/\mu = 0.01$.

Working

TOPIC 6 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

TOPIC 7

Gauss's, Green's and Stokes' Theorems and Applications; Single-Valued and Multivalued Functions

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

The integral theorems of vector calculus

These three theorems link an integral over a region to an integral over its boundary. They generalise the Fundamental Theorem of Calculus.

Theorem	Statement	Relates
Green's theorem (plane)	$\oint_C (P \, dx + Q \, dy) = \iint_R (\partial Q / \partial x - \partial P / \partial y) \, dA$	line integral $\leftrightarrow$ double integral
Stokes' theorem	$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$	circulation $\leftrightarrow$ flux of the curl
Gauss's (divergence) theorem	$\oiint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{F} \, dV$	outward flux $\leftrightarrow$ volume integral of divergence

Applications include computing areas with line integrals ($A = \frac{1}{2} \oint (x \, dy - y \, dx)$), deriving the continuity equation and the heat-conduction equation, Gauss's and Ampère's laws in electromagnetism, and checking whether a field is conservative.

Single-valued and multivalued functions

A **single-valued function** gives exactly one output for each input. Examples are polynomials and e^z . A **multivalued function** gives more than one output for some inputs:

- ▶ $z^{1/2}$ has two values, and $z^{1/n}$ has n values;
- ▶ $\log z = \ln|z| + i(\arg z + 2k\pi)$ has infinitely many values;
- ▶ the inverse trigonometric functions are multivalued.

To work with them we choose a **principal branch**, place a **branch cut** (a curve across which the function jumps), and identify **branch points**, where going round a small circle changes the value. For example, $z = 0$ is a branch point of $\sqrt{z}$ and of $\log z$. Getting branches right is essential for the contour integration in Topic 8 and for physical problems with slits, cracks and flows around corners.

Application questions

7.01
AGE

A field boundary is traced by GPS as the ellipse $x = 300 \cos t$, $y = 200 \sin t$ (m). Use Green's theorem ($A = \frac{1}{2} \oint (x \, dy - y \, dx)$) to find the field area. Explain how a GPS area-measuring device can use the same formula on a list of boundary points (the shoelace formula).

Working

7.02
AGE

The groundwater flux under a farm is $\mathbf{q} = (x^2, y^2, z^2)$ (m/day, scaled) in the cube $0 \leq x, y, z \leq 1$. Check the divergence theorem by computing the net outward flux both as a surface integral and as a volume integral.

Working

7.03
CHE

Apply Gauss's divergence theorem to a fixed control volume V to derive the continuity equation $\partial\rho/\partial t + \nabla \cdot (\rho\mathbf{v}) = 0$ from the integral mass balance.

Working

7.04
CHE

The complex potential of flow in a corner-shaped mixing channel involves $w = z^{2/3}$. Find all values of w at $z = 8i$, give the principal value, and state the branch point. Suggest a suitable branch cut for a channel occupying the upper half-plane.

Working

7.05

C P E

Polygon-area routines in computer graphics are based on Green's theorem. Derive the shoelace formula $A = \frac{1}{2} \sum (x_i y_{i+1} - x_{i+1} y_i)$ from $\frac{1}{2} \oint (x \, dy - y \, dx)$, and use it on the polygon with vertices $(0, 0)$, $(4, 0)$, $(4, 3)$, $(2, 5)$ and $(0, 3)$. Write the algorithm in pseudocode.

Working

7.06

C P E

A function $\text{atan2}(y, x)$ returns the principal argument of $z = x + iy$ in $(-\pi, \pi]$. Explain why $\arg z$ is multivalued. Show what happens to the computed angle as a point travels once round the origin, and describe how "phase unwrapping" algorithms fix the resulting jump.

Working

7.07
C V E

Use Green's theorem to find the area and centroid of the plane region bounded by $y = x^2$ and $y = 2x$ (in metres), which models a curved slab section. Check the answer with direct double integration.

Working

7.08
C V E

Seepage under a dam has flux field $\mathbf{q} = (2x - y, x + 3y, -5z)$ (per unit time). Check whether the field is consistent with an incompressible soil ($\nabla \cdot \mathbf{q} = 0$). Use the divergence theorem to find the net outflow from the unit cube.

Working

7.09
EEE

Starting from Ampère's circuital law $\oint_C \mathbf{H} \cdot d\mathbf{r} = I_{\text{enc}}$, use Stokes' theorem to derive the differential form $\nabla \times \mathbf{H} = \mathbf{J}$. Apply it to find $\mathbf{H}$ inside a long straight conductor of radius a carrying a uniformly distributed current I .

Working

7.10
EEE

The phase of an AC signal is found from the complex logarithm of the phasor. Find all values of $\log(-1 + i\sqrt{3})$, give the principal value, and explain what the choice of branch means when measuring phase angles above 180° .

Working

7.11

FDE

A food can is heated in a retort. Heat flux into the can is $\mathbf{q} = -k\nabla T$. Use the divergence theorem on the can volume to show that the total rate of heat entering through the surface equals $\iiint \rho c_p (\partial T / \partial t) dV$ (when there is no internal generation). This is the basis of the heat-conduction equation.

Working

7.12

FDE

The stirring flow in a mixing bowl is approximated by $\mathbf{v} = (-y, x, 0)$ (rad/s units). Use Stokes' theorem to compute the circulation around the bowl rim (a circle of radius 0.15 m in the plane $z = 0.1$ m) through the flat disc it bounds. Check the answer against a direct line integral.

Working

7.13
MEE

Compute the work done by the force $\mathbf{F} = (y + z, z + x, x + y)$ round the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ using Stokes' theorem. Explain the result in terms of conservative forces.

Working

7.14
MEE

The flow round a sharp re-entrant corner of angle $3\pi/2$ in a duct is modelled by $w = z^{2/3}$. Identify the branch point, describe the branch cut, and find the two other values of w at $z = 1$ besides the principal value. Explain why the velocity becomes unbounded at the corner.

Working

7.15
PTE

Use the divergence theorem to show that for steady incompressible flow in a reservoir with no wells ($\nabla \cdot \mathbf{u} = 0$), the net volumetric flux through any closed surface is zero. Extend the argument to a region containing a producing well of rate q , and derive the resulting flux condition.

Working

7.16
PTE

The complex potential for flow towards a well at the origin is $w = (q/(2\pi)) \log z$. Show that the stream function $\psi = \text{Im } w$ is multivalued, find its change on going once round the well, and relate this jump to the well's production rate.

Working

TOPIC 7 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS

T O P I C 8

Analytic Functions, Cauchy–Riemann Equations, Singularities and Zeros, Contour Integration and Bilinear Transformation

16 questions · 2 per department · Read the key ideas first, then answer your department's questions before the others.

KEY IDEAS

Complex analysis studies functions $f(z) = u(x, y) + i v(x, y)$ of the complex variable $z = x + iy$.

- ▶ **Analytic (holomorphic) functions:** f is analytic at z_0 if it is differentiable in a neighbourhood of z_0 . Analytic functions are very well behaved. They are infinitely differentiable and can be expanded in power series.
- ▶ **Cauchy–Riemann equations:** a necessary condition for analyticity is $u_x = v_y$ and $u_y = -v_x$. If the partial derivatives are also continuous, the condition is sufficient too. As a result, u and v are **harmonic** ($\nabla^2 u = \nabla^2 v = 0$), and v is the *harmonic conjugate* of u . This is why complex analysis can model 2-D potential flow, electrostatics and steady heat conduction.
- ▶ **Zeros and singularities:** z_0 is a zero of order m if $f(z_0) = \dots = f^{(m-1)}(z_0) = 0$ but $f^{(m)}(z_0) \neq 0$. **Singularities** are points where f is not analytic. They are classified, using the Laurent series, as *removable*, *poles* (of order m) or *essential*.
- ▶ **Contour integration:**
 - *Cauchy's integral theorem:* $\oint_C f(z) dz = 0$ if f is analytic on and inside the closed curve C .
 - *Cauchy's integral formula:* $f(z_0) = (1/2\pi i) \oint_C f(z)/(z - z_0) dz$, and its extension to derivatives.
 - *Residue theorem:* $\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$. This is used to evaluate real integrals and inverse Laplace transforms.
- ▶ **Bilinear (Möbius) transformation:** $w = (az + b)/(cz + d)$ with $ad - bc \neq 0$. It is conformal (it preserves angles), maps circles and lines to circles and lines, and is fixed uniquely by three point correspondences. It is used to map awkward domains onto simple ones, and underlies the Smith chart.

Complex analysis is essential in control-system stability (Nyquist criterion), AC circuit analysis, signal processing (the z -transform), aerodynamics (the Joukowski aerofoil) and potential-flow problems.

Application questions

8.01 AGE

Steady groundwater flow under a drainage field is modelled by the potential $\phi(x, y) = x^2 - y^2$. Check that ϕ is harmonic. Use the Cauchy–Riemann equations to find the stream function ψ and the complex potential $w = \phi + i\psi$, and sketch the flow lines. Where is the stagnation point?

Working

8.02
AGE

The transfer function of a greenhouse temperature controller is $G(s) = 10/((s + 1)(s + 2)(s + 5))$. Find its poles and zeros, and use the residue theorem to find the impulse response $g(t)$, the inverse Laplace transform, as $(1/2\pi i) \oint e^{st} G(s) ds$.

Working

8.03

CHE

Use the Cauchy–Riemann equations to show that $f(z) = \bar{z} = x - iy$ is *not* analytic anywhere, while $f(z) = e^x(\cos y + i \sin y)$ is analytic everywhere. Identify the second function as e^z . Explain why analyticity is needed when the concentration field in a thin 2-D film is modelled as the real part of an analytic function.

Working

8.04

CHE

The response of a first-order-plus-lag process is $C(s) = 1/(s(\tau s + 1)^2)$. Classify its singularities, and use residues to find $c(t)$ for $\tau = 2$ min.

Working

8.05

C P E

A digital filter has transfer function $H(z) = (z^2 - 1)/(z^2 - 0.5z + 0.06)$. Find its zeros and poles, and state whether the filter is stable (all poles inside $|z| = 1$). Write a program sketch that computes the poles and tests stability for any given denominator coefficients.

Working

8.06

C P E

The inverse z-transform is $x[n] = (1/2\pi i) \oint X(z) z^{n-1} dz$. Use the residue theorem to find $x[n]$ for $X(z) = z/((z - 0.5)(z - 0.25))$ with $|z| > 0.5$.

Working

8.07

C V E

Seepage under a sheet pile is modelled by the complex potential $w = \cosh^{-1}(z/b)$. Rewrite it as $z = b \cosh w$, separate the real and imaginary parts, and show that the equipotential lines are ellipses and the streamlines are hyperbolae. Identify the branch points at $z = \pm b$.

Working

8.08

C V E

Evaluate the real integral $\int_{-\infty}^{\infty} dx/(x^2 + 4)^2$ by contour integration over a semicircle in the upper half-plane. The integral appears in the deflection of a beam on an elastic foundation under a distributed load.

Working

8.09

EEE

Find the bilinear transformation that maps $z = 0, 1, \infty$ to $w = -1, 0, 1$. Show that the normalised load-impedance plane ($\operatorname{Re} z \geq 0$) is mapped onto the unit disc $|w| \leq 1$, and explain how this gives the Smith chart reflection coefficient $\Gamma = (z - 1)/(z + 1)$.

Working

8.10

EEE

An RLC circuit has transfer function $H(s) = 1/(s^2 + 2s + 5)$. Find its poles, and classify them as simple or of higher order. Use Cauchy's integral formula / residues to find the impulse response $h(t)$.

Working

8.11
FDE

The temperature in a long rectangular food slab in steady state is $T(x, y) = 20 + 5(x^3 - 3xy^2)$. Show that T is harmonic, and find the harmonic conjugate that gives the heat-flow lines. Write the analytic function $f(z)$ whose real part is $T - 20$.

Working

8.12
FDE

Evaluate $\oint_C (e^{2z}/(z-1)^3) dz$ where C is $|z| = 2$, using Cauchy's integral formula for derivatives. Such integrals arise in inverting Laplace-domain models of thermal processing with repeated time constants.

Working

8.13

MEE

The Joukowski transformation $w = z + 1/z$ maps circles in the z -plane to aerofoil-like shapes. Find the singularities and the critical points (where $dw/dz = 0$) of the transformation, and show that the circle $|z| = 1$ maps to the segment $[-2, 2]$ on the real axis.

Working

8.14

MEE

Evaluate $\int_0^{2\pi} d\theta/(5 + 4 \cos \theta)$ by contour integration round the unit circle. This type of integral appears in the analysis of rotating machinery with eccentric loading.

Working

8.15

PTE

A source (injector) at $z = -a$ and a sink (producer) at $z = a$ give the complex potential $w = (q/2\pi)[\log(z + a) - \log(z - a)]$. Find the singularities of w and their type, and show that the equipotentials are circles (Apollonius circles). Find the stagnation points, if there are any.

Working

8.16

PTE

Find the bilinear transformation that maps the circle $|z| = R$ (the reservoir boundary) to the unit circle, with an off-centre well at $z = z_0$ going to the centre $w = 0$. Explain how this lets the known solution for a centred well be used for an off-centre well.

Working

TOPIC 8 REVIEW

How confident do I feel? ☐ Not yet ☐ Getting there ☐ Confident ☐ Could teach it

Questions to ask my lecturer or tutor:

Tutor check: _____

Date: _____

THE TOPIC IN MY OWN WORDS